

CRITICAL

Phishing Attempt with Malicious URL

Source: Proofpoint | Created: Dec 26, 2025, 03:13 PM GMT+5:30 | MITRE: T1566



SCAN TO VERIFY

FINAL VERDICT

True Positive - Confirmed Threat

INVESTIGATION SCORE



RAW LOG

```
{
  "timestamp": "2023-10-01T09:15:30Z",
  "event_type": "email_received",
  "src_ip": "198.51.100.22",
  "email_sender": "noreply@secure-bank.com",
  "url": "http://secure-bank.com/login",
  "username": "target_user",
  "hostname": "HQ-DNS-349",
  "dst_ip": "10.0.10.18"
}
```

EVIDENCE COLLECTED

4 indicators

TYPE	VALUE	OSINT VERDICT	CRIT
IP	198.51.100.22	MALICIOUS	YES
IP	10.0.10.18	CONFIRMED	-
URL	http://secure-bank.com/login	MALICIOUS	YES
EMAIL	noreply@secure-bank.com	SUSPICIOUS	-

CONTAINMENT ACTIONS

- [+] Block Ip
- [+] Isolate Host
- [+] Reset Credentials
- [+] Block Hash
- [+] Collect Forensics

AI ANALYSIS

AI ANALYSIS FEEDBACK

Verdict is correct (+40). All critical artifacts identified (+30). Response actions correct (+30). Speed Bonus: Solved quickly (+10)

EDUCATIONAL

EDUCATIONAL WALKTHROUGH

AI MENTOR

Investigation Walkthrough

Reading the Logs

First, let's focus on the following fields in the raw log:

1. `src_ip: `198.51.100.22`` - This IP address is crucial as it is the source of the email. Analyzing this IP through OSINT tools can reveal its reputation and history of malicious activity.
2. `email_sender: `noreply@secure-bank.com`` - While this appears to be a legitimate sender, we must verify the domain and its association with phishing attempts.
3. `url: `http://secure-bank.com/login`` - The URL is a critical artifact. We need to check if this URL has been flagged for phishing or malicious activities.

Why These Artifacts Matter

- **IP Address (`198.51.100.22`)**: This IP has been reported 500 times for hosting phishing campaigns. This high volume of reports indicates a strong likelihood of malicious intent. An analyst should recognize that even if the email appears legitimate, the source IP's reputation is a significant red flag.
- **URL (`http://secure-bank.com/login`)**: This URL is associated with phishing attempts. Even though it resembles a legitimate banking URL, the context of its use in this alert suggests it could be a spoofed site designed to capture user credentials. Analysts should be cautious about URLs that mimic trusted domains.
- **Email Sender (`noreply@secure-bank.com`)**: While this email address might seem legitimate, it is marked as "critical: false" because the sender's domain can be spoofed. Analysts should not rely solely on the sender's address without corroborating evidence from other artifacts.

Verdict Reasoning

The verdict of true positive is supported by the following chain of evidence:

1. The source IP has a well-documented history of phishing activity, which is a strong indicator of malicious intent.
2. The URL is flagged in OSINT databases for phishing, reinforcing the suspicion raised by the source IP.
3. Although the email sender appears legitimate, it does not outweigh the evidence provided by the IP and URL. If the IP had a clean reputation or if the URL had no prior reports, we might consider this a false positive.

A false positive could occur if the IP had a clean history or if the URL was verified as legitimate through multiple trusted sources. Analysts must weigh all evidence carefully.

Why These Response Actions

- **block_ip**: This action is necessary because the source IP is known for hosting phishing campaigns. Blocking it prevents further malicious communications from this source.
- **block_url**: Similar to blocking the IP, this action prevents users from accessing the phishing site, protecting them from potential credential theft.
- **reset_credentials**: This is a proactive measure to safeguard the target user. If they have interacted with the phishing attempt, their credentials may be compromised. Resetting their credentials helps mitigate the risk of unauthorized access.

... (truncated — full walkthrough at infoseclabs.io)